

THE SACRED SCRIPTS

Synthetic Data Suite: Backend Operations

I. The Awakening Protocol

To begin the rites of generation, one must first summon the necessary spirits of the machine.

Step 1: The Messenger (Redis)

The "Waiting Room" must be open for communication to flow.

- **Rite:** Ensure Docker is active.
- **Incantation:** Run the following in your terminal:

```
docker run -d -p 6379:6379 redis:alpine
```

If the machine speaks of "Conflict," the Messenger already walks among us.

Step 2: The Brain (FastAPI)

This handles the logical records and the gateway for your commands.

```
cd backend
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process
.\venv\Scripts\activate
uvicorn app.main:app --reload
```

Verification: Seek the Oracle at <http://127.0.0.1:8000/docs>.

Step 3: The Muscle (Celery Worker)

The force that manifests images from the void.

```
cd backend
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process
.\venv\Scripts\activate
$env:PYTHONPATH="."; celery -A app.tasks.celery_worker worker --
loglevel=info -P solo
```

Verification: Await the signal: "celery@computername ready".

II. The Deep Cleansing

Should the connections falter, these rituals of reset shall restore order.

```
# To halt Redis:
docker stop $(docker ps -q)

# To clear Port 8000:
Stop-Process -Id (Get-NetTCPConnection -LocalPort 8000).OwningProcess -
Force
```

III. The Cartography of Data

- **Great Repository:** E:\synthetic-data-suite\backend\synthetic_data.db
- **Scroll Image Vault:** E:\synthetic-data-suite\backend\uploads\
- **Heavy Artifacts:** Stored within the D: Drive.